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EXPERIMENT 26

Prolog Program: Fruits and Their Colours Using Backtracking

Aim

To write a Prolog program to store fruits and their colours and retrieve the fruit-colour relationships using **backtracking**.

Algorithm

1. Start the Prolog program.
2. Define facts representing each fruit and its colour.
3. Create a rule to find the colour of a given fruit.
4. Load the program in GNU Prolog.
5. Query the fruit colour.
6. Use ; after an answer to activate **backtracking** and find the next possible answer.
7. Display the results.

Code

```
% Fruits and their colours


fruit_colour(apple, red).

fruit_colour(banana, yellow).

fruit_colour(orange, orange).

fruit_colour(grapes, green).

fruit_colour(mango, yellow).
```

fruit_colour(guava, green).

fruit_colour(watermelon, green).

fruit_colour(strawberry, red).

% Rule to find the colour of a fruit

fruit(X, Y) :-

    fruit_colour(X, Y).

Input / Queries

fruit(apple, Colour).

Output

```
| ?- consult('C:/Users/MOSHIYA/Downloads/exp 26.pl').
compiling C:/Users/MOSHIYA/Downloads/exp 26.pl for byte code...
C:/Users/MOSHIYA/Downloads/exp 26.pl compiled, 14 lines read - 1138 bytes written, 16 ms

yes
| ?- fruit(apple, Colour).

Colour = red

yes
| ?-
```

Result

Thus, the Prolog program to represent fruits and their colours using backtracking was successfully implemented and executed. By using ;, Prolog backtracks through the facts and displays the next possible fruit-colour combination.